Lab 1

Math 452_552

08/28/2026

Objectives

- Learn the basic Jupyter Notebook workflow.
- Load data into a pandas DataFrame.
- Estimate simple probabilities from real data.
- Create a simple graph.

Verify setup

1. Open a Jupyter notebook using the shortcut you created.
2. In a blank Python cell, run the following code to prove to me that your Python is working.

```
import sys
import os

print(sys.executable)
print(os.getcwd())
```

3. Load the General Social Survey data and display the first five observations.

Before loading the data, check (from above) your working directory. If your working directory is `Math 452_552` use:

```
import pandas as pd

gss = pd.read_csv("data/gss_bayes.csv", index_col=0)
gss.head()
```

Otherwise, if your notebook is saved in a subfolder such as Lab or Labs, your working directory should be that folder so use:

```
import pandas as pd

gss = pd.read_csv("../data/gss_bayes.csv", index_col=0)
gss.head()
```

The `../` means go up one directory before looking for the data folder.

4. Run the pandas command `gss.info()`. Explain the output.

Data Exploration

Run each command in a separate Python cell.

```
gss.columns
gss.shape
```

```
gss.describe()
gss['sex'].value_counts()
gss['age'].describe()
```

1. How many variables are there?
2. What does `describe()` tell you?
3. Is sex numeric or categorical? Explain.
4. What does `value_counts()` do?
5. Run the command `gss.isna().sum()`. Which variables contain missing values?

Probability Estimation

- Estimate the probability that a randomly selected respondent is female.

```
p_female = (gss['sex'] == 2).mean()
print(p_female)
```

Why does `.mean()` compute a probability?

- Estimate the probability a respondent is over age 50.
- Estimate the probability that a randomly selected respondent is female **and** over age 50.
- What is the probability that someone is over 50 **given** they are female? (Hint: this is a conditional probability)
- Is the previous probability the same as the probability that someone is female **given** they are over 50? Explain.
- What is the probability that someone is over 50 given they are male?

Plotting

- Create a histogram of age.

```
import matplotlib.pyplot as plt

gss["age"].hist(bins=20)

plt.xlabel("Age")
plt.ylabel("Frequency")
plt.title("Distribution of Respondent Age")
plt.show()
```

- Describe the shape of the age distribution.
- Is it symmetric? Skewed? Are there any unusual features?

Reflection

In two or three sentences, describe one thing you learned about Jupyter Notebook and one thing you learned about estimating probabilities from data.

Deliverable

After completing this lab, generate a .pdf file and upload it to the dropbox on D2L.